

POPULATION BASED TRAINING OF NEURAL NETWORKS & BAYESIAN OPTIMIZATION : INTRODUCTION

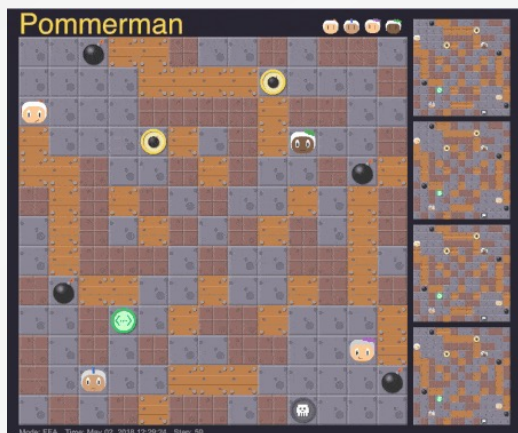
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Introduction

- Neural networks dominate the modern machine learning landscape, but their training and success still suffer from sensitivity to empirical choices of hyperparameters. **Model architecture, loss function, optimisation algorithm** etc.
- Population Based Training (PBT), a simple asynchronous optimisation algorithm which effectively utilises a fixed computational budget to jointly optimise a population of models and their hyperparameters to maximise performance (**Hyperparameter schedules**).

Applications

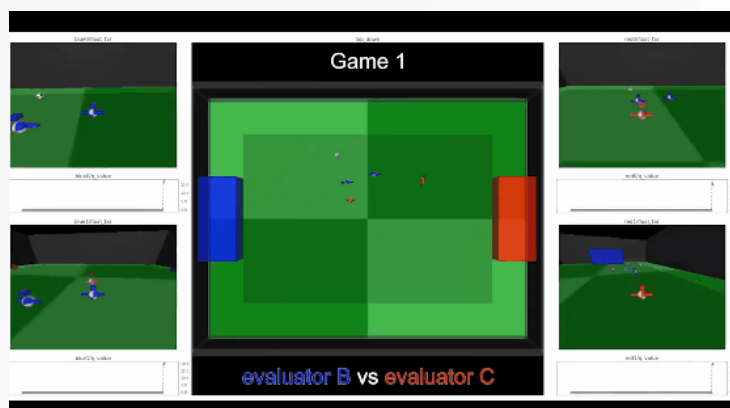


The NeurIPS 2018 Pommernan Competition

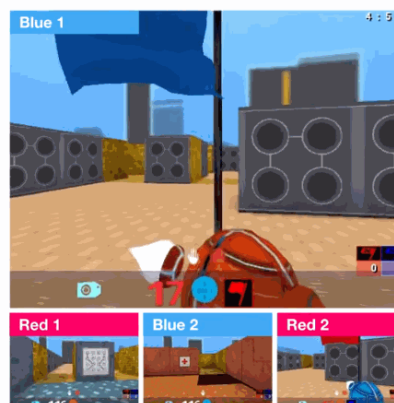


StarCraft II

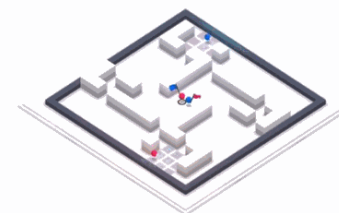
Agent observation raw pixels



Soccer



Quake III Arena Capture the Flag



Indoor map overview

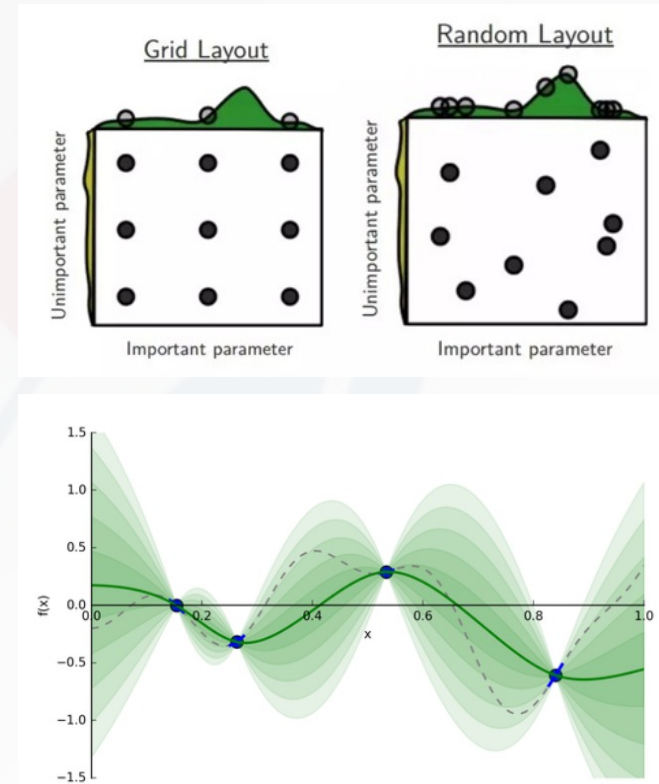
Related Work

- Parallel search

Performs many parallel optimisation processes, each with different hyperparameters, with a view to finding a single best output from one of the optimisation processes – examples of this are **grid search** and **random search**.

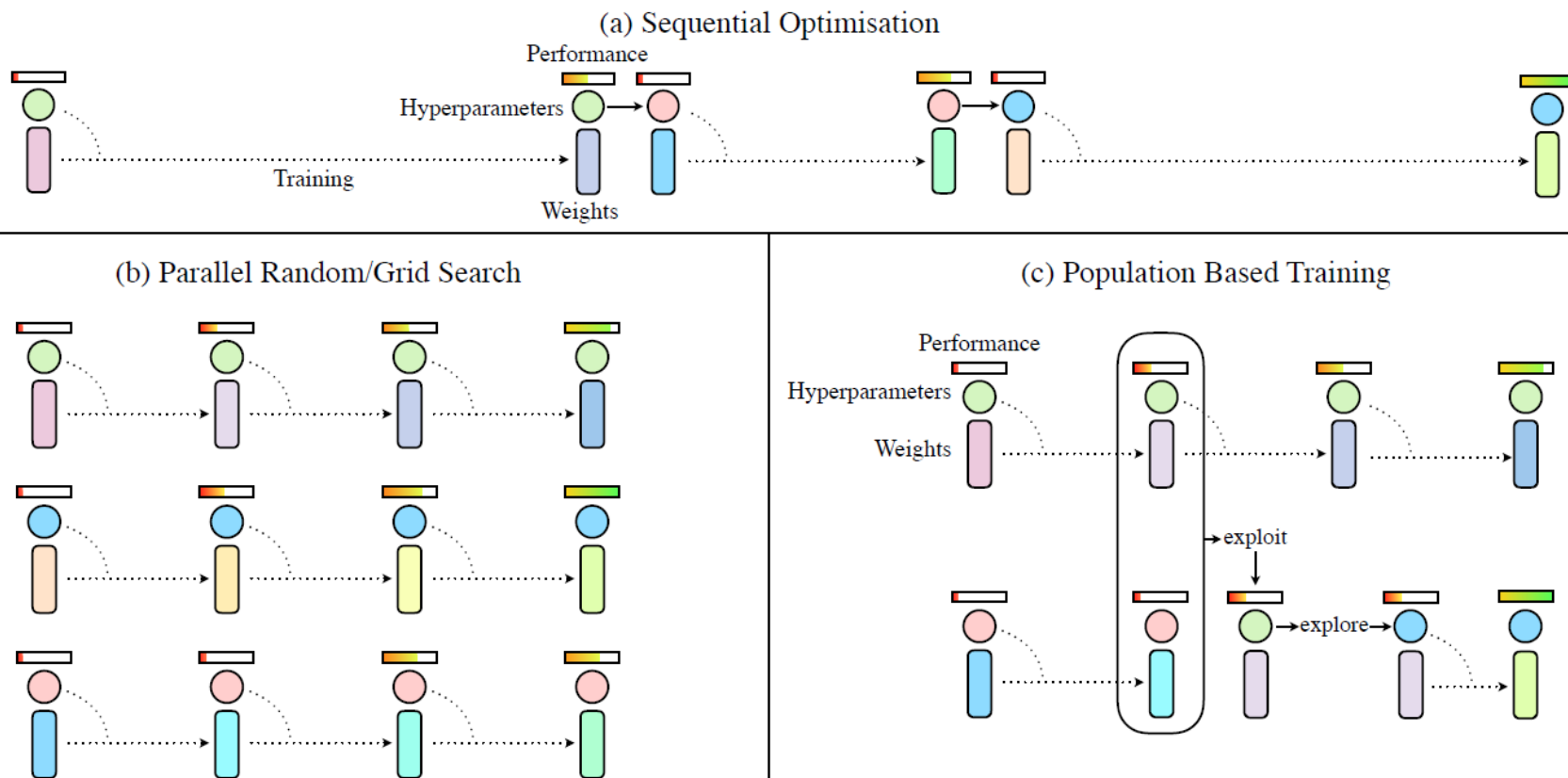
- Sequential optimization

Sequential optimization performs few optimisation processes in parallel, but does so many times sequentially, to gradually perform hyperparameter optimisation using information obtained from earlier training runs to inform later ones – examples of this are **hand tuning** and **Bayesian optimisation**.



- Population Based Training (PBT) bridges and extends parallel search methods and sequential optimisation methods.
 - (a) automatic selection of hyperparameters during training.
 - (b) online model selection to maximise the use of computation spent on promising models.
 - (c) the ability for online adaptation of hyperparameters to enable non-stationary training regimes and the discovery of complex hyperparameter schedules.

Population Based Training

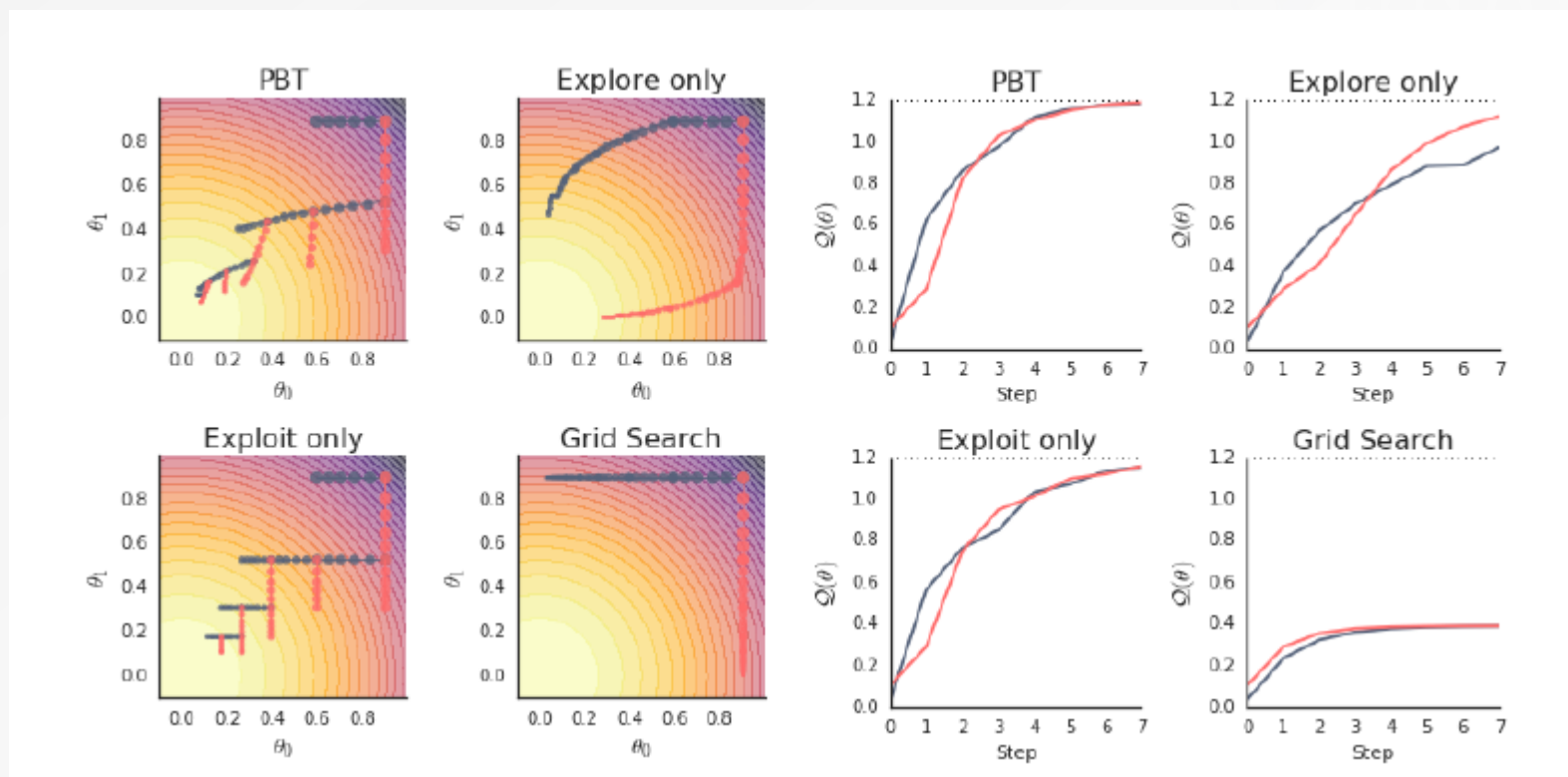


Pseudocode

Algorithm 1 Population Based Training (PBT)

```
1: procedure TRAIN( $\mathcal{P}$ ) ▷ initial population  $\mathcal{P}$ 
2:   for  $(\theta, h, p, t) \in \mathcal{P}$  (asynchronously in parallel) do
3:     while not end of training do (parameter, hyperparameter, evaluator, step)
4:        $\theta \leftarrow \text{step}(\theta|h)$  ▷ one step of optimisation using hyperparameters  $h$ 
5:        $p \leftarrow \text{eval}(\theta)$  parameter update ▷ current model evaluation
6:       if ready( $p, t, \mathcal{P}$ ) then
7:          $h', \theta' \leftarrow \text{exploit}(h, \theta, p, \mathcal{P})$  ▷ use the rest of population to find better solution
8:         if  $\theta \neq \theta'$  then
9:            $h, \theta \leftarrow \text{explore}(h', \theta', \mathcal{P})$  ▷ produce new hyperparameters  $h$ 
10:           $p \leftarrow \text{eval}(\theta)$  perturbing or resampling ▷ new model evaluation
11:        end if
12:      end if
13:      update  $\mathcal{P}$  with new  $(\theta, h, p, t + 1)$  ▷ update population
14:    end while
15:  end for
16:  return  $\theta$  with the highest  $p$  in  $\mathcal{P}$ 
17: end procedure
```

Toy Example



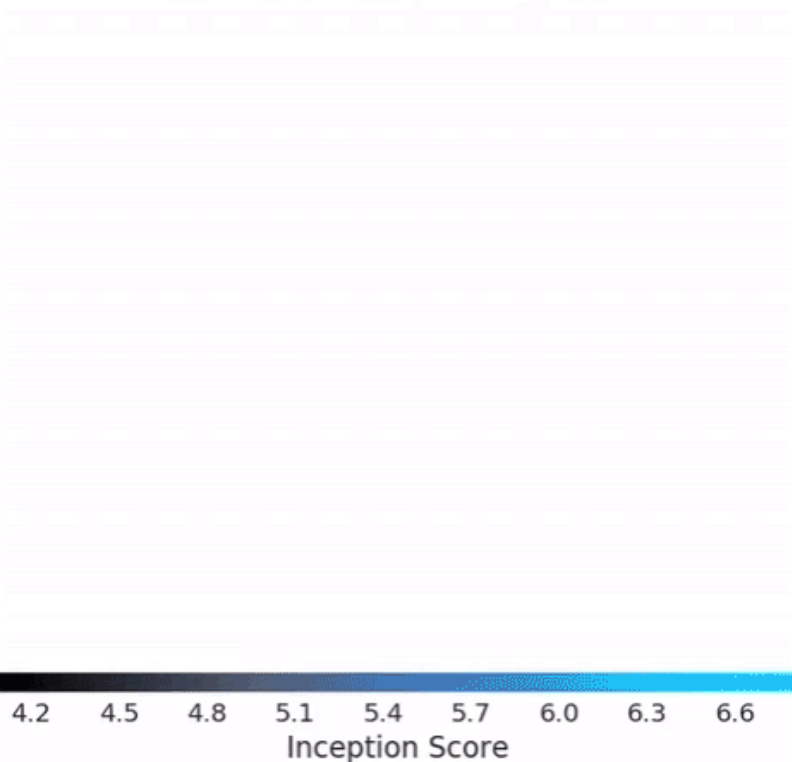
maximise *objective function*: $Q(\theta) = 1.2 - (\theta_0^2 + \theta_1^2)$

surrogate function: $\hat{Q}(\theta|h) = 1.2 - (h_0\theta_0^2 + h_1\theta_1^2)$

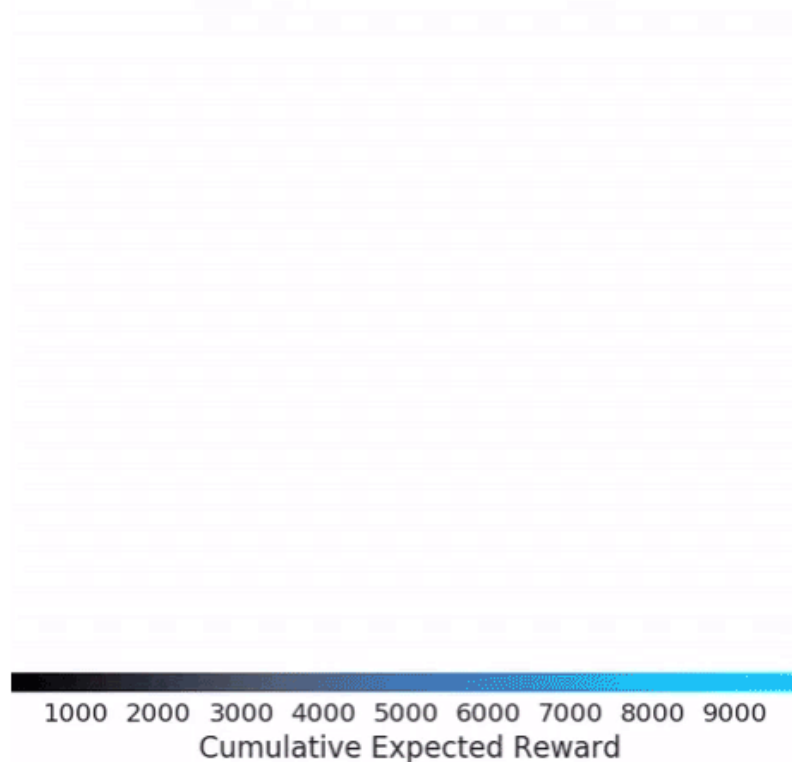
initial $\theta = [0.9, 0.9]$ *hyperparameter* $h = [1, 0]$ & $[0, 1]$

Visualisation of PBT

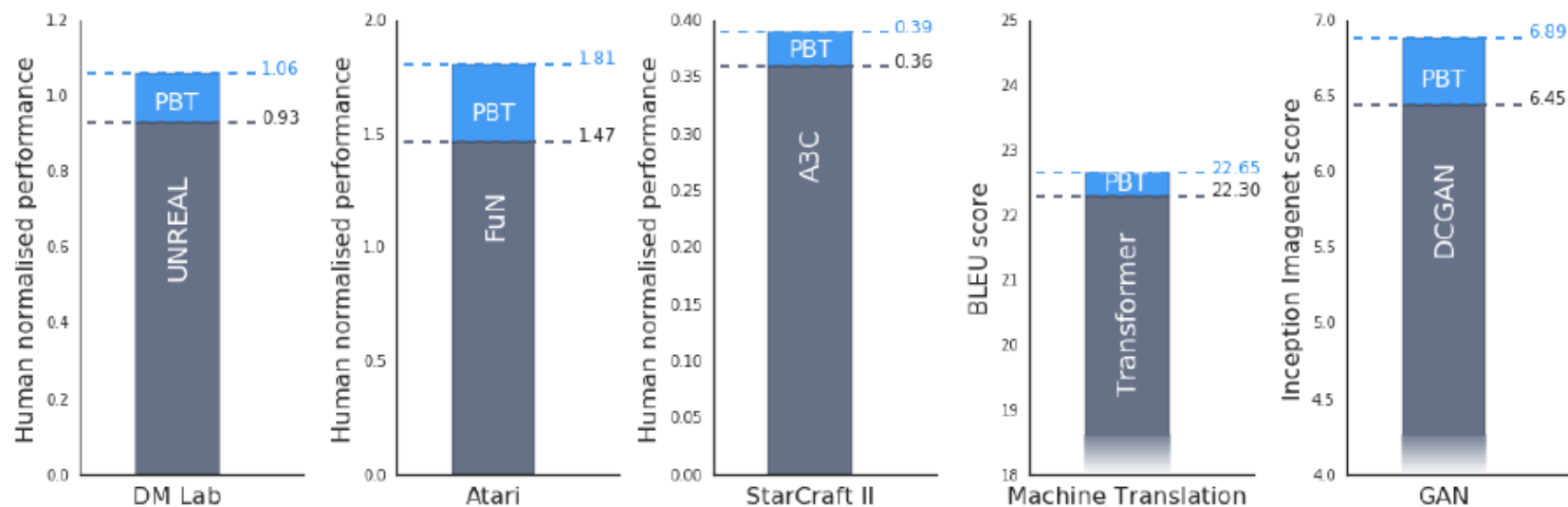
GAN population development



FuN population development



Results



Workers:

40

80

30

32

45

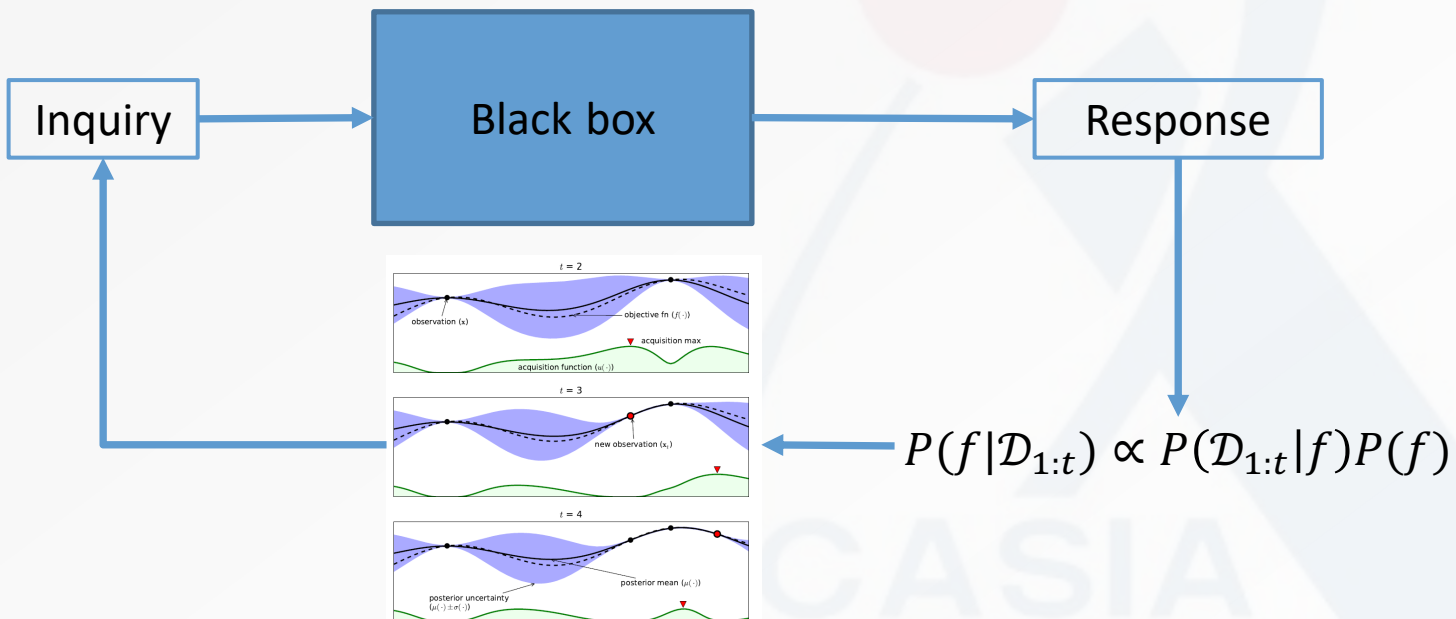
Bayesian Optimization

- Optimizing a nonlinear function $f(\mathbf{x})$ over a compact set $\mathcal{A}$,

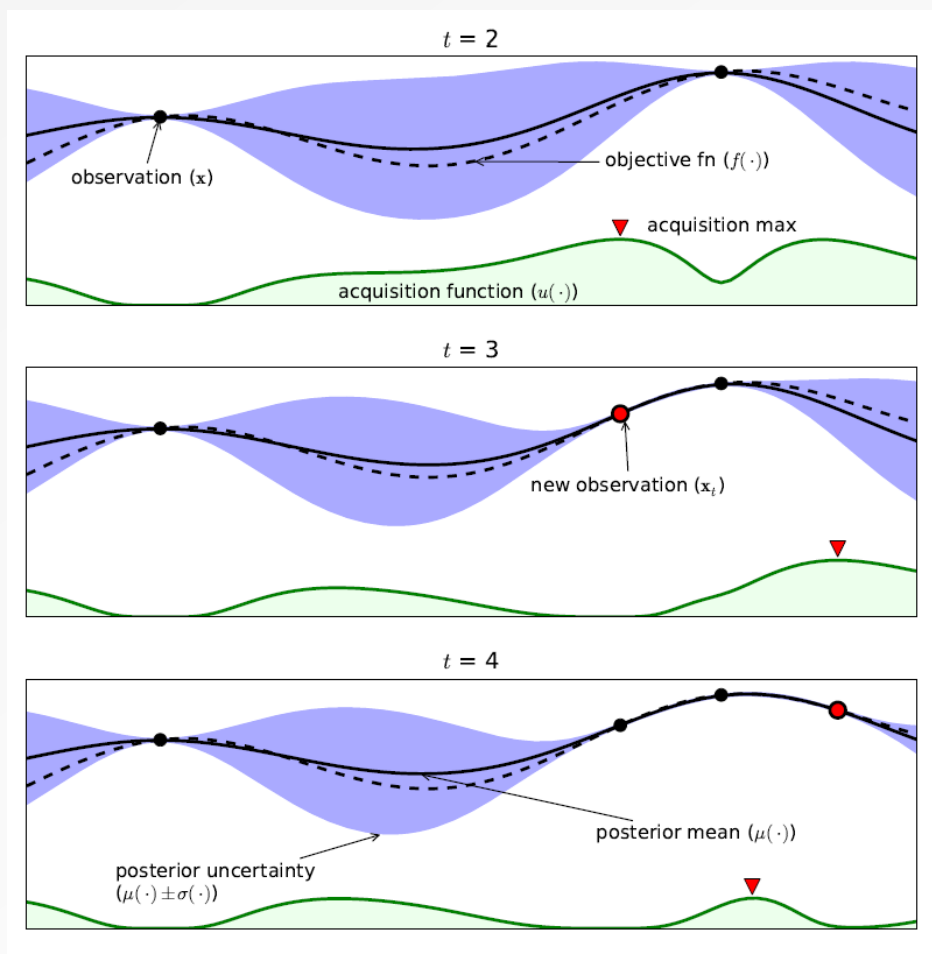
$$\max_{\mathbf{x} \in \mathcal{A} \subset \mathbb{R}^d} f(\mathbf{x})$$

- We assume that the objective function is Lipschitz-continuous.
That is, there exists some constant C , such that for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{A}$:

$$\|f(\mathbf{x}_1) - f(\mathbf{x}_2)\| \leq C \|\mathbf{x}_1 - \mathbf{x}_2\|$$



Toy Example



Priors over functions

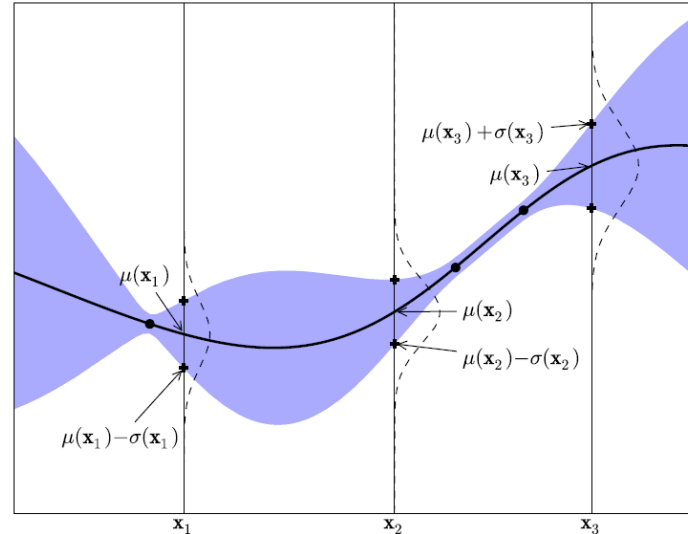
- Gaussian process

$$f(x) \sim \mathcal{GP}((m(x), k(x, x')))$$

- covariance function k

$$k(x_i, x_j) = e^{-\frac{1}{2}\|x_i - x_j\|^2}$$

$$\mathbf{K} = \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_t) \\ \vdots & \ddots & \vdots \\ k(x_t, x_1) & \cdots & k(x_t, x_t) \end{bmatrix}$$



- Let us denote the value of the function at this arbitrary point as $f_{t+1} = f(x_{t+1})$. Then, by the properties of Gaussian processes, $\mathbf{f}_{1:t}$ and f_{t+1} are jointly Gaussian:

$$\begin{bmatrix} \mathbf{f}_{1:t} \\ f_{t+1} \end{bmatrix} \sim \mathcal{N}(\mathbf{0}, \begin{bmatrix} \mathbf{K} & \mathbf{k} \\ \mathbf{k}^T & k(x_{t+1}, x_{t+1}) \end{bmatrix})$$

- Using the Sherman-Morrison-Woodbury formula

$$P(f_{t+1} | \mathcal{D}_{1:t}, x_{t+1}) = \mathcal{N}(\mu_t(x_{t+1}), \sigma_t^2(x_{t+1}))$$

- where

$$\mu_t(x_{t+1}) = \mathbf{k}^T \mathbf{K}^{-1} \mathbf{f}_{1:t} \quad \sigma_t^2(x_{t+1}) = k(x_{t+1}, x_{t+1}) - \mathbf{k}^T \mathbf{K}^{-1} \mathbf{k}$$

Acquisition Functions

PI

- maximizing the probability of improvement over the incumbent $f(x^+)$, where $x^+ = \operatorname{argmax}_{x_i \in x_{1:t}} f(x_i)$

$$PI(x) = P(f(x) \geq f(x^+)) = \Phi\left(\frac{\mu(x) - f(x^+)}{\sigma(x)}\right)$$

$$PI(x) = P(f(x) \geq f(x^+) + \xi) = \Phi\left(\frac{\mu(x) - f(x^+) - \xi}{\sigma(x)}\right), \xi \geq 0$$

EI

- defined the improvement function as:

$$I(x) = \max\{0, f_{t+1}(x) - f(x^+)\}$$

- The new query point is found by maximizing the expected improvement

$$x = \operatorname{argmax}_x E(\max\{0, f_{t+1}(x) - f(x^+)\} | \mathcal{D}_t)$$

Acquisition Functions

- The likelihood of improvement I on a normal posterior distribution characterized by $\mu(x), \sigma^2(x)$ can be computed from the normal density function,

$$\frac{1}{\sqrt{2\pi}\sigma(x)} e^{-\frac{(\mu(x)-f(x^+)-I)^2}{2\sigma^2(x)}}$$

- The expected improvement is the integral over this function:

$$\begin{aligned} E(I) &= \int_{I=0}^{I=\infty} I \frac{1}{\sqrt{2\pi}\sigma(x)} e^{-\frac{(\mu(x)-f(x^+)-I)^2}{2\sigma^2(x)}} dI \\ &= \sigma(x) \left[\frac{\mu(x)-f(x^+)}{\sigma(x)} \Phi\left(\frac{\mu(x)-f(x^+)}{\sigma(x)}\right) + \phi\left(\frac{\mu(x)-f(x^+)}{\sigma(x)}\right) \right] \end{aligned}$$

Acquisition Functions

- The expected improvement can be evaluated analytically, yielding:

$$E(I) = \begin{cases} (\mu(x) - f(x^+))\Phi(Z) + \sigma(x)\phi(Z), & \text{if } \sigma(x) > 0 \\ 0, & \text{if } \sigma(x) = 0 \end{cases}$$

$$Z = \frac{\mu(x) - f(x^+)}{\sigma(x)}$$

$$E(I) = \begin{cases} (\mu(x) - f(x^+) - \xi)\Phi(Z) + \sigma(x)\phi(Z), & \text{if } \sigma(x) > 0 \\ 0, & \text{if } \sigma(x) = 0 \end{cases}$$

$$Z = \begin{cases} \frac{\mu(x) - f(x^+) - \xi}{\sigma(x)}, & \text{if } \sigma(x) > 0 \\ 0, & \text{if } \sigma(x) = 0 \end{cases}$$

Acquisition Functions

- Confidence bound criteria

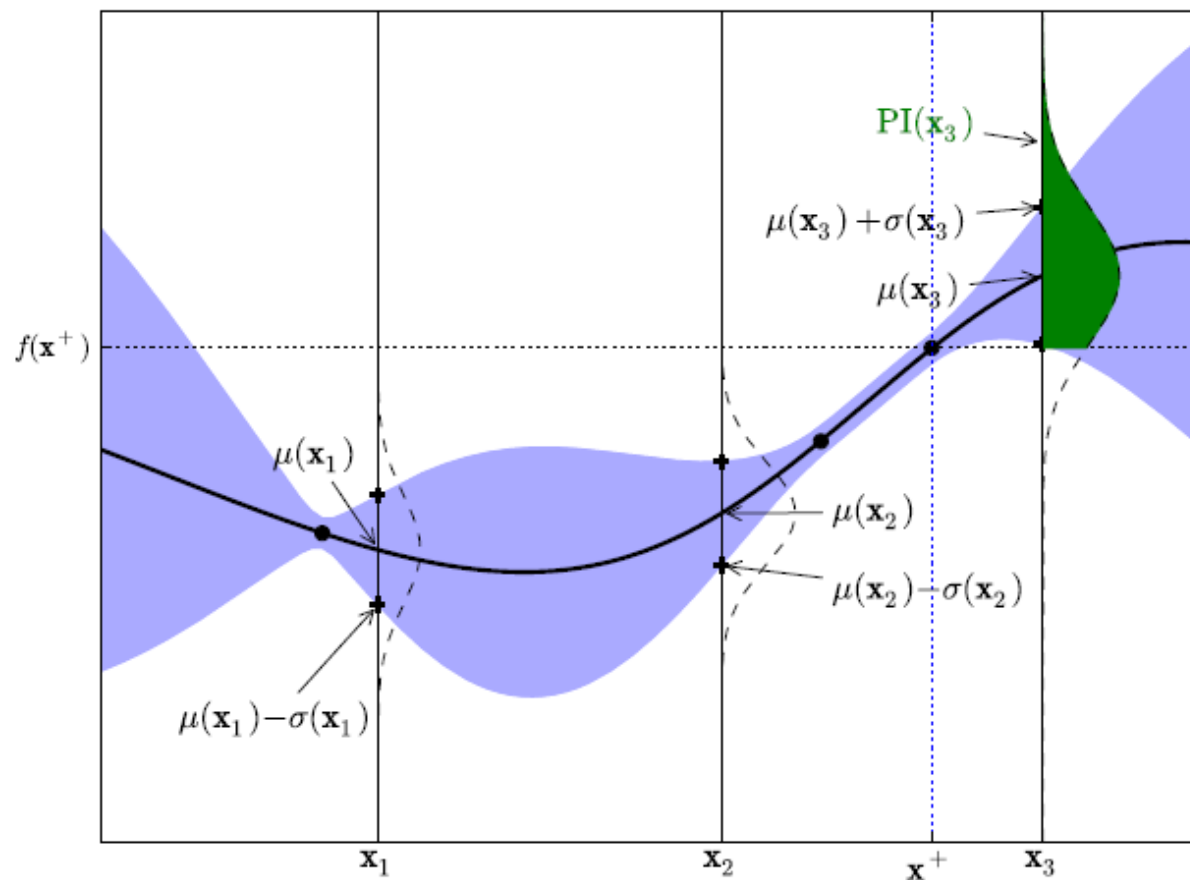
lower confidence bound:

$$LCB(x) = \mu(x) - \kappa\sigma(x), \kappa \geq 0$$

upper confidence bound:

$$UCB(x) = \mu(x) + \kappa\sigma(x), \kappa \geq 0$$

Acquisition Functions





Thanks!

